ADITHYA B K

AI/ML Engineer | Applied Machine Learning & NLP

+91 8971095311 | adithya.bk01@gmail.com

linkedin.com/in/adithyabk01
github.com/Adithyabk01
new-portfolio-ruby-nu-50.vercel.app
Mandya, India

Computer Science undergraduate specializing in Artificial Intelligence & Machine Learning, with hands-on experience across the ML lifecycle — data preprocessing, model training and evaluation, NLP, and LLM integration (Llama3, Gemini). Built applied AI products in healthcare, agriculture, infrastructure and fintech, and turned data into decision-ready insights through analytics and dashboards.

EDUCATION

PES College of Engineering, Mandya Oct 2023 – Oct 2027

B.E. in Computer Science and Engineering — Artificial Intelligence & Machine Learning, 7.73 CGPA (through 6th semester)

EXPERIENCE

Machine Learning Trainee — Internship, Employability.life Aug 2024 – Oct 2024

- Built a supervised-learning cancer-cell classification pipeline on a synthetic dataset, covering preprocessing, model training, and evaluation.
- Applied correlation analysis to reduce feature dimensionality from 32 to 18 features; evaluated performance using 5-fold cross-validation.
- Technologies: Python, Scikit-learn, Pandas.

PROJECTS

MediRead — AI-Based Medical Report Understanding System — NLP & LLM Integration

- Built a pipeline to extract lab values from PDF medical reports using text parsing and regex, with rule-based abnormal-value detection and concern classification.
- Integrated Llama3 (via Ollama) to generate multilingual plain-language explanations with voice output; deployed a web version on Vercel using Hugging Face inference.

Tech Stack: Python, NLP, Llama3/Ollama, Hugging Face, React, Vercel

SmartPave — AI-Based Pavement Maintenance Optimization — Predictive Modeling

- Trained a scikit-learn Random Forest regressor (300 trees, median imputation) to predict pavement condition (PCI) across 12 sections of the NH-275 corridor; reported R^2, MAE, RMSE and feature importance from the model.
- Added life-cycle-cost present-value analysis, a rule-based maintenance matrix and a FastAPI + React dashboard for decision support.

Tech Stack: Python, scikit-learn, Pandas, NumPy, FastAPI, React, TypeScript

Kisan Sethu — AI-Powered AgriTech Marketplace — AI Features & Full-Stack

- Integrated Gemini Vision to grade crop quality from photos (grade, score, shelf life, defects) with a fallback path, plus demand/price forecasting charts and a multilingual voice-onboarding flow.
- Built a FastAPI backend with JWT authentication and role-based access across four user types, an autonomous reverse-auction bidding agent, and a reputation/credit scoring system.

Tech Stack: Python, FastAPI, Gemini API, Next.js, TypeScript, SQLAlchemy

RecoverAI — AI Revenue Recovery Agent (Razorpay Buildathon, Track 03) — Agent Design

- Designed a 6-stage pipeline (risk detection, failure diagnosis, AI recommendation, 9-rule policy and safety gate, simulation, audit trail) over a 1,000-record synthetic payment dataset.
- Separated AI recommendation from authorization so no action runs without passing deterministic safety checks; every decision is logged with an audit ID.

Tech Stack: TypeScript, React, Tailwind CSS

Startup Investment Trends in India — Data Analytics & BI

- Cleaned and structured startup funding data with Python and MySQL, and built interactive Power BI dashboards showing sector-wise and city-wise investment trends.

Tech Stack: Python, MySQL, Power BI

TECHNICAL SKILLS

ML & Data: Scikit-learn, Pandas, NumPy, Random Forest, EDA, Feature Selection, Cross-Validation, Data Cleaning
NLP & LLMs: NLP (TF-IDF), LangChain, Llama3 (Ollama), Gemini API, Hugging Face, Prompt Engineering
Languages: Python, SQL, Java, C++, C, TypeScript, Swift
Backend & Databases: FastAPI, REST APIs, MySQL, PostgreSQL, SQLite, Power BI
Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Linux, Vercel

CERTIFICATIONS

- Machine Learning with Python — IBM
- Fundamentals of NLP — University of Colorado Boulder
- Introduction to AI and Machine Learning on Google Cloud — Google Cloud
- Deloitte Australia Data Analytics Job Simulation — Forage
- Prompt Engineering for ChatGPT — Vanderbilt

ACHIEVEMENTS

- Qualified for Smart India Hackathon (SIH) 2025 at the national level.
- Runner-Up, Minute to Win (ISTE) — placed 2nd of 50+ participants.
- Industry 4.0 Foundation Certification — Microsoft & SAP CSR Initiative (TechSaksham / Edunet Foundation / AICTE).